ModelIt!

Teacher's Guide

Why Do Bridges Collapse?

Modeling Structural Forces, Material Fatigue, and Engineering Failure

Quick Reference

Grade Level	Grade 11L1
NGSS Standard	HS-PS2-1, HS-ETS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS2-1, HS-ETS1-2

Analyze data to support the claim that Newton's second law of motion describes the mathematical relationship among the net force on a macroscopic object, its mass, and its acceleration. Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students analyze force distribution data and fatigue accumulation curves to identify failure risks and predict the remaining safe service life of bridge structures.
Disciplinary Core Idea	**PS2.A Forces and Motion / ETS1.C Optimizing the Design Solution** — Forces act on bridge structures through tension, compression, and shear. Engineering solutions must be optimized iteratively, balancing safety margins against cost and practical constraints.
Crosscutting Concept	**Stability and Change** — Students analyze how bridges appear stable while undergoing gradual internal change through fatigue damage, and identify the threshold where gradual change produces sudden catastrophic instability.

CAST Testing Alignment

MULTIPLE CHOICE:

A steel truss bridge built in 1975 with a Factor of Safety of 2.0 carries an average of 20,000 vehicles per day. Inspections were conducted every 2 years until 2005, then deferred due to budget cuts. In 2025, the bridge is 50 years old with no inspection in 20 years. Based on your model, what is the most likely current condition?

CONSTRUCTED RESPONSE:

Using your model, explain why the I-35W bridge in Minneapolis collapsed in 2007 despite carrying normal traffic loads that day. Describe the role of material fatigue, deferred maintenance, and the relationship between cumulative damage and catastrophic failure. Include model data showing how the bridge's Structural Integrity likely declined over its 40-year lifespan.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

On August 1, 2007, the I-35W bridge in Minneapolis collapsed during rush hour, dropping 111 vehicles into the Mississippi River and killing 13 people. The bridge was not struck by a tornado or earthquake — it was carrying normal traffic on an ordinary Wednesday. The collapse was caused by decades of material fatigue, corrosion, and deferred maintenance that had invisibly reduced the bridge's structural capacity below its design safety threshold. Understanding how and why bridges fail requires modeling the invisible process of material fatigue.

Forces in Bridges: Tension, Compression, and Shear

Every bridge must transfer the weight of vehicles and its own structure safely to the ground through a series of structural members under different types of stress. Tension stretches members (like cables in a suspension bridge), compression squeezes them (like columns and piers), and shear acts perpendicular to the member's length (like the force on a beam at its supports). Different bridge designs distribute these forces differently: arches convert loads primarily to compression, trusses use a network of triangles to convert loads to axial forces, and suspension bridges hang the deck from cables in tension. Engineers analyze these forces using Newton's laws and statics to ensure every member can safely carry its share.

Material Fatigue: The Silent Killer

When a material is subjected to a single large force, it either holds or breaks — this is ultimate strength. But material fatigue is different: it occurs when forces well below ultimate strength are applied repeatedly. Each load cycle creates microscopic damage at stress concentration points — holes, welds, sharp corners, and surface scratches. Over millions of cycles, a tiny crack initiates and grows incrementally with each cycle. The crack is invisible without specialized testing (ultrasonic, radiographic, magnetic particle) until it reaches a critical length, at which point the remaining material can no longer support the load and fracture occurs suddenly. Fatigue causes an estimated 90% of all metallic structural failures.

The I-35W Bridge: Anatomy of a Collapse

The I-35W bridge was a steel deck truss bridge built in 1967. A 2001 study classified it as 'structurally deficient' due to fatigue cracking and corrosion. In 2005, it was rated '4 out of 9' for structural condition — meaning 'poor.' Despite these warnings, the bridge was kept in service. On the day of the collapse, construction work had placed additional steel and aggregate on the deck, adding approximately 262 tons of extra dead load. The NTSB investigation found that undersized gusset plates (connecting plates at truss joints) had corroded and fatigued over 40 years, and the additional construction load pushed these weakened plates past their reduced capacity. The bridge did not fail because of extraordinary circumstances — it failed because ordinary circumstances met a structure that had been silently weakening for decades.

Infrastructure Crisis: The Scope of the Problem

The American Society of Civil Engineers (ASCE) gives US infrastructure a grade of C- overall, with bridges specifically rated C. As of 2023, approximately 42% of all US bridges are over 50 years old, and 7.5% (approximately 46,100 bridges) are classified as structurally deficient. The backlog of bridge maintenance and repair is estimated at $125 billion. State departments of transportation lack the funding to inspect and maintain all bridges at recommended intervals, forcing risk-based triage. The consequence is that millions of Americans cross bridges every day that are carrying invisible fatigue damage at unknown levels — a situation that transforms a manageable engineering problem into a public safety crisis.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify applied load, structural design type, maintenance interval, force distribution, fatigue damage, and structural integrity as the key components of the bridge safety system.

E **Establish Relationships**

Students determine that applied load, structural design type, and maintenance interval are external factors controlled by traffic management, engineering design, and policy decisions, while force distribution, fatigue damage, and structural integrity are internal physical outcomes determined by material properties and loading history.

V **Visualize & Model**

Students build a computational model that simulates force distribution through a bridge design and tracks fatigue damage accumulation over a multi-decade service life.

E **Evaluate Outcomes**

Students run normal service life, deferred maintenance, and overloading scenarios to observe how fatigue accumulates over time and how maintenance frequency affects the timeline to potential failure.

R **Revise & Extend**

Students add corrosion rate, temperature cycling, or redundancy level to explore how environmental factors and design choices affect long-term structural reliability.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"You drove over a bridge today. Or yesterday. Probably without thinking about it. But 42% of American bridges are over 50 years old, and 46,000 are classified as structurally deficient. How do you know the one you crossed is safe?"*

Watch For

Show the I-35W bridge collapse photos — before, during, and after. Ask: What do you think caused this? Was it one event or something that built up over time?

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling the invisible process that kills bridges — material fatigue — and discovering why maintenance is literally a matter of life and death."*

Watch For

Have students read objectives. Pre-teach 'material fatigue' and 'factor of safety' with a paper clip demonstration (bend it back and forth until it breaks).

Slide 3: Big Question

3 min

Say: *"Engineers can calculate exactly how strong a bridge is. So why do bridges still collapse? The answer is time — and something happening inside the steel that no one can see."*

Watch For

Demonstrate fatigue with a paper clip: first bend doesn't break it, neither does the second, but eventually it snaps. Ask: Which bend broke it? (All of them did.)

Slide 4: LEVER Framework

2 min

Say: *"We're going to model a bridge's entire life — from day one to potential collapse — and find out what determines whether it lasts 50 years or fails catastrophically."*

Watch For

Preview LEVER steps. Emphasize that this model runs over DECADES to reveal invisible damage accumulation.

Slide 5: Activity 1: Components

8 min

Say: *"What determines whether a bridge stays standing? Some factors are choices we make, others are physics we can't control."*

Watch For

Guide sorting: Applied Load, Structural Design Type, and Maintenance Interval are external. Force Distribution, Fatigue Damage, and Structural Integrity are internal physical outcomes.

Slide 6: Activity 2: Connections

8 min

Say: *"Every truck is a tiny hammer blow to the bridge's steel. One blow does nothing. A million do nothing visible. But at 10 million, cracks form. At 100 million, the cracks are growing. Where does it end?"*

Watch For

Students map the fatigue accumulation pathway. Key insight: fatigue damage is cumulative and irreversible — you can't 'un-fatigue' a bridge, only detect and repair cracks.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's run two versions of the same bridge for 50 years: one with regular inspections and one where the government cuts the maintenance budget. Predict which one is still standing in 2075."*

Watch For

Students predict Structural Integrity curves for both scenarios. Run simulations. The maintained bridge holds; the deferred one approaches failure. Test the overloading event on both.

Slide 8: Discoveries

5 min

Say: *"Here's the terrifying truth: a bridge can look perfectly fine and be one load cycle away from catastrophic failure. Fatigue is invisible without inspection. That's why maintenance isn't optional — it's survival."*

Watch For

Show the paper clip analogy one more time. Then show real fatigue crack photos from bridge inspections. Discuss: Why do governments cut maintenance budgets when the consequences are this severe?

Slide 9: STEM Challenge

12 min

Say: *"You're the chief engineer for a state with 1,200 bridges, $50 million in maintenance budget, and 40% of your bridges over 50 years old. Not enough money for everything. What do you prioritize — and who might die if you get it wrong?"*

Watch For

Groups design risk-based maintenance plans allocating budget across bridge portfolios. Present with model-based risk rankings and inspection schedules.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Applied Load, Structural Design Type, Maintenance Interval
- **INTERNAL:** Force Distribution, Fatigue Damage, Structural Integrity

Relationships

- **Applied Load to Force Distribution:** DETERMINES — The magnitude and position of loads on the bridge determine how forces are distributed through structural members. Heavier concentrated loads create higher stress at supports and connections. The structural design type determines how those forces are channeled — trusses distribute to axial forces in members, arches convert to compression, beams experience bending.
- **Force Distribution to Fatigue Damage:** POSITIVE (+) — Higher forces at stress concentration points (welds, holes, connections) drive faster fatigue crack growth. Each load cycle advances existing cracks by a microscopic amount proportional to the stress intensity at the crack tip. Poorly distributed forces that create stress concentrations accelerate fatigue damage significantly.
- **Maintenance Interval to Structural Integrity:** POSITIVE (+) INDIRECT — More frequent maintenance does not prevent fatigue damage — it detects it early. When inspectors find cracks below critical length, they can be repaired by welding, grinding, or adding reinforcement. Regular maintenance maintains Structural Integrity by catching damage before it becomes dangerous. Deferred maintenance allows cracks to grow unchecked toward catastrophic failure.

Key Discoveries

- Material fatigue is invisible and cumulative — a bridge can lose 30% of its structural integrity over decades while appearing perfectly fine to visual inspection, because fatigue cracks grow inside the metal at the microscopic level
- Regular maintenance and inspection extend bridge life dramatically — not because they repair fatigue damage, but because they detect cracks early when they can be reinforced, before reaching critical length for sudden fracture
- Deferred maintenance creates a dangerous illusion of safety — the bridge looks the same on the outside, but internal fatigue damage is accumulating unchecked, and the margin between current load and failure capacity is shrinking invisibly
- Most bridge collapses are not caused by a single extraordinary event but by the combination of cumulative fatigue, deferred maintenance, and a load event that would have been survivable on a well-maintained structure

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Bridges collapse because of design or construction errors"	While design and construction defects can cause failures, the majority of bridge collapses are caused by material fatigue accumulated over decades of normal service, compounded by insufficient maintenance and inspection. The bridge was designed correctly for its expected loads — but fatigue is a time-dependent process that was poorly understood when many current bridges were built, and maintenance has not kept pace with the aging infrastructure. The model shows that even a perfectly designed bridge will eventually fail if fatigue damage is not detected and repaired.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If a bridge hasn't collapsed yet, it must be safe"	This is survivorship bias applied to infrastructure. A bridge that has stood for 50 years may have accumulated significant invisible fatigue damage and may be much closer to failure than anyone realizes. The I-35W bridge stood for 40 years before collapsing during normal traffic. The model shows that Structural Integrity can decrease by 40% without any visible external signs. 'It hasn't failed yet' is not evidence of safety — it's evidence that the failure threshold hasn't been crossed YET.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We can see when a bridge is deteriorating — it will show signs of damage"	Surface corrosion and visible cracks are obvious, but the most dangerous type of damage — internal fatigue cracking — is completely invisible without specialized testing equipment. Fatigue cracks initiate at microscopic defects inside the metal and grow internally for years before reaching the surface. By the time a fatigue crack is visible, it may already be at or near critical failure length. This is why non-destructive testing methods (ultrasonic, radiographic, magnetic particle) are essential — and why visual-only inspections are insufficient for detecting fatigue.